Marcelo Lares | Lead Data Scientist | Applied AI | Machine Learning Engineering

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Applied AI and Data Science technical leader with experience leading production ML initiatives, defining experimentation strategies, mentoring scientists, and coordinating engineering, product, analytics, and client stakeholders. Specialized in ranking models, statistical learning, feature engineering, and end-to-end model validation at LATAM scale. Former CONICET researcher and Associate Professor with a PhD in Astronomy, 30+ Q1 publications, and experience supervising doctoral researchers.

Technical Leadership

AI Strategy: Led Data Science initiatives from problem framing to production delivery, balancing model quality, business impact, cost, and operational risk.

Roadmaps & OKRs: Defined technical and product roadmaps, contributed to OKRs, tracked execution, and aligned delivery with KPIs.

Experimentation: Designed production experiment strategies to maximize learning while controlling exposure, cost, and deployment risk.

Cross-functional Work: Coordinated Data Science, engineering, product, analytics, and client stakeholders across academic and industry settings.

Talent & Evaluation: Mentored junior data scientists, onboarded team members, supervised doctoral researchers, evaluated technical work, and participated in hiring decisions.

Core Expertise

Machine Learning: PyTorch, Scikit-Learn, XGBoost, ranking, recommender systems, embeddings, feature engineering, computer vision

Production ML: ONNX, MLflow, feature stores, Fury pipelines, CUDA, A/B testing, Datadog, Looker

Data & AI: Python, SQL, R, NumPy, Pandas, SciPy, BigQuery, LangChain, RAG, prompt engineering

Engineering: Git, GitHub, Linux, APIs, Sphinx, AWS (S3/EC2), GCP (BigQuery/Cloud Storage)

Domains: E-commerce, retail, public revenue, tourism, agriculture, international trade

Professional Experience

Mercado Libre

Senior Data Scientist / Machine Learning Engineer, Argentina *2024–2026*

- Served as the primary data science owner for a production online ranking model powering recommendations for dozens of internal clients and millions of users across Argentina, Brazil, and Mexico.
- Guided production experimentation decisions for ranking models, recommending when to launch, pause, or refine experiments based on statistical evidence, business KPIs, cost, and operational risk.
- Developed and maintained the PyTorch ranking model, adding features, modifying embeddings, and integrating production feature-store data.
- Adapted Fury pipelines and optimized framework and CUDA configurations, tripling the amount of training data the model could process.
- Prepared ONNX artifacts, validated production configurations, and partnered with engineering to ensure reliable production deployments.
- Implemented MLflow experiment tracking and designed a pre-production validation protocol for safer, reproducible model releases.
- Developed statistical tools to quantify confidence in A/B-test differences and monitored clicks, purchases, GMV, and model behavior through Datadog and Looker.
- Communicated trade-offs, risks, and next steps to engineering, product, analytics, and leadership stakeholders, shaping model-release decisions.
- Delivered measurable lift in purchase conversion through controlled production experiments while coordinating engineering, product, and analytics teams.

IThreex Global

Lead Data Scientist, Argentina *2022–2024*

- Led an eight-person data science team delivering client-facing AI and analytics solutions across public revenue, retail, tourism, agriculture, and international trade.
- Defined the technical roadmap for the Molibdeno AI platform, prioritizing reusable workflows, client needs, and delivery constraints.
- Defined OKRs and tracked team execution toward client-facing AI and analytics deliverables.
- Designed and implemented the core Python library used by the Molibdeno AI platform to standardize reusable machine-learning workflows.
- Served as the principal technical contact for Kolektor, developing payment-behavior analyses and tax-revenue forecasts for large taxpayers in Córdoba.
- Delivered segmentation and predictive solutions that supported marketing and commercial strategies for Pueblo Nativo, Inverfin, and Agencia ProCórdoba.
- Developed computer-vision models for image-based cattle-weight estimation and defined the corresponding image-acquisition protocols.
- Built a LangChain-based RAG support service exposed through an API.
- Presented technical progress and results to clients and stakeholders, translating business requirements into data products.
- Led, mentored, and onboarded data scientists, assigning technical direction across client projects.
- Participated in candidate interviews, technical hiring decisions, performance evaluation, and difficult staffing decisions.

Academic and Research Background

Universidad Nacional de Córdoba

Associate Professor, Argentina *2023–present*

Head of the Data Science course in the Applied Mathematics department. Twenty years teaching statistics, machine learning, and scientific computing while supervising graduate students and developing technical talent.

CONICET

Researcher, Argentina *2004–2023*

Applied Bayesian inference, statistical learning, numerical methods, and HPC to large astronomical datasets. Developed open-source scientific software and automated analysis pipelines, collaborated in international multidisciplinary teams, supervised researchers, reviewed scientific work, and organized academic activities.

Education and Selected Achievements

Universidad Nacional de Córdoba

PhD in Astronomy, Argentina *2009*

Dissertation focused on statistical analysis of large-scale astronomical datasets.

Research: Author of 30+ peer-reviewed articles in international Q1 journals and multiple open-source scientific software projects; h-index 17.

Mentorship: Supervised two PhD researchers and one graduate researcher; mentored junior scientists and data scientists across academic and industry settings.

Technical Evaluation: Served as peer reviewer for international journals, evaluated research work and academic candidates, and participated in data science hiring decisions.

Profiles: Google Scholar ORCID GitHub LinkedIn